
Timer
-timeval start_time
-timeval end_time
+void start()
+double get_elapsed_seconds()
BruteForce
-const TspGraph& graph_ref
-vector<int> s
-int num_elements
-double best_cost
+BruteForce(const TspGraph& graph)
+void run(int num_cities)
+double get_best_cost()
-bool perm1()
-void swap(int p, int q)
-void evaluate_cost()
GeneticAlgorithm
-const TspGraph& graph_ref
-int cities
-double best_cost
-mt19937 rng
+GeneticAlgorithm(const TspGraph& graph)
+void run(int num_cities, int pop_size, int num_generations, int mutation_percent)
+double get_best_cost()
-void evaluate_population(vector<Tour>& population)
-void mutate(Tour& tour)
-Tour crossover(const Tour& parent1, const Tour& parent2)
-Tour generate_random_tour()
references
references
manages
0..*
TspGraph
+static constexpr int max_cities = 20
-array<double, 400> distance_matrix
+TspGraph()
+void load_distances(string file_path)
+double get_distance(int city_a, int city_b)
«struct»
Tour
+vector<int> path
+double cost
+bool operator<(const Tour& other)